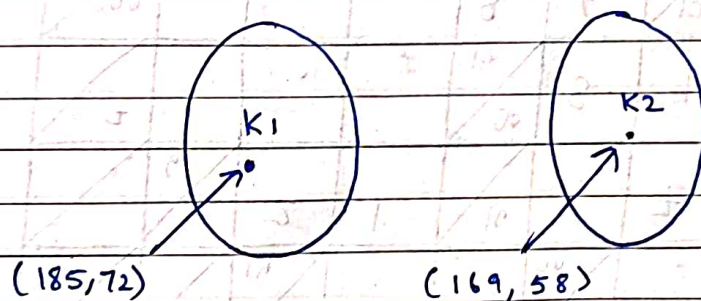


## \* UNSUPERVISED LEARNING

### \* K-Means Algorithm

|   | Height | Weight |
|---|--------|--------|
| ① | 185    | 72     |
| ② | 170    | 56     |
| ③ | 168    | 60     |
| ④ | 179    | 68     |
| ⑤ | 182    | 72     |
| ⑥ | 188    | 77     |
| ⑦ | 180    | 71     |
| ⑧ | 180    | 70     |
| ⑨ | 183    | 84     |
| ⑩ | 180    | 88     |
| ⑪ | 180    | 67     |
|   | 177    | 76     |



$$\text{E.D for } ③ \quad K_1 = \sqrt{(168 - 185)^2 + (60 - 72)^2}$$
$$= 20.80$$

$$K_2 = \sqrt{(168 - 169)^2 + (60 - 58)^2}$$
$$= 4.48$$

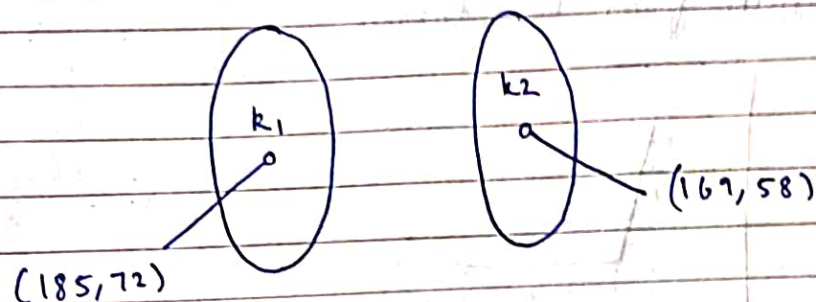
Hence ③ belongs to  $K_2$

$$K_1 = \{1\}$$

$$K_2 = \{2, 3\}$$

As in  $k_2$  we have 2 points now  
 Hence for  $k_2$  we need to calculate new centroid

$$\text{for } k_2 = \left( \frac{170+168}{2}, \frac{56+60}{2} \right) = (169, 58)$$



E.D for ④  $k_1 = \sqrt{(179-185)^2 + (68-72)^2} = 6.32$

$$k_2 = \sqrt{(179-169)^2 + (68-58)^2} = 14.14$$

Hence 4 belongs to  $k_1$

$$k_1 = \{1, 4\}$$

$$k_2 = \{2, 3\}$$

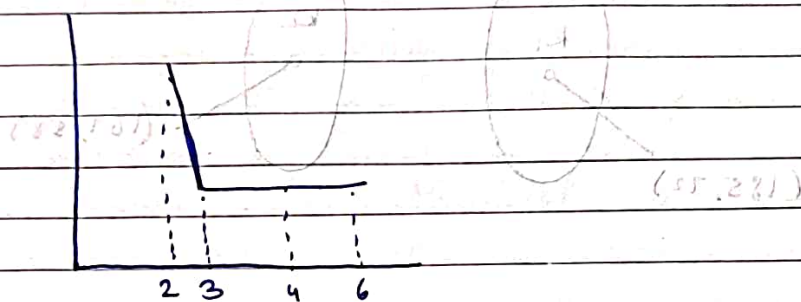
Go on like this.....

$$k_1 = \{1, 4, 5, 6, 7, 8, 9, 10, 11, 12\}$$

$$k_2 = \{2, 3\}$$

① In ML journey wherever you find K - that means euclidean distance matrix

② How to find how many clusters I want to make.  
elbow method or elbow graph will decide how many cluster we need to make from the dataset



③ In K-means - x-axis is number of clusters! (K) say 4-7  
y-axis is WCSS within cluster sum of squares  
 $P_1 \cdot P_1 = (22-22)^2 + (11-11)^2 = 0$

$$WCSS = \sum_{P_i \text{ in cluster } 1} \text{distance}(P_i, (1))^2 + \sum_{P_i \text{ in cluster } 2} \text{distance}(P_i, (2))^2 +$$

$$\sum_{P_i \text{ in cluster } 3} \text{distance}(P_i, (3))^2$$

